

WEST PLAINS, MO, USA

WMO: 723484

Lat: 36.878N

Lon: 91.903W

Elev: 1223

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53901

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	14.6	-1.8	5.2	14.8	2.6	6.6	17.8	22.1	39.2	19.8	39.6	6.8	330	0.443

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.5	94.6	74.6	91.4	74.4	89.5	74.2	78.0	89.3	76.9	87.6	75.6	85.5	6.7	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.7	135.9	84.0	73.3	129.7	82.3	72.5	126.2	81.3	42.3	89.1	41.0	87.5	39.9	86.0	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.9	16.8	15.1	DB	2.5	98.1	4.1	4.5	-0.4	101.4	-2.8	104.0	-5.1	106.6	-8.1	109.9	(4)
(5)			WB	1.4	79.7	3.7	1.2	-1.3	80.6	-3.5	81.3	-5.6	82.0	-8.3	82.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.3	34.7	38.1	48.0	57.8	65.8	74.2	77.9	77.0	69.5	58.2	47.2	37.4
	DBStd	17.44	11.14	11.19	10.52	8.42	7.45	5.26	5.05	5.30	6.80	8.73	9.39	9.86
	HDD50	1628	487	358	163	28	2	0	0	0	0	25	160	405
	HDD65	4221	939	752	531	240	85	4	0	1	32	246	535	856
	CDD50	4276	13	26	102	261	493	726	864	835	586	279	77	14
	CDD65	1394	0	0	5	23	110	280	399	372	168	35	2	0
	CDH74	13200	0	3	43	207	827	2600	4059	3624	1535	287	15	0
	CDH80	5149	0	0	4	26	174	939	1811	1617	527	51	0	0
Wind	WSAvg	6.7	7.7	7.9	8.3	8.3	6.7	5.6	4.9	4.8	5.0	6.1	7.1	7.5
Precipitation	PrecAvg	45.8	3.0	3.0	4.4	5.0	5.2	3.7	3.9	2.9	4.3	3.7	4.2	3.1
	PrecMax	56.0	5.6	7.4	13.7	9.5	11.9	6.6	7.0	5.9	12.2	12.0	8.8	6.8
	PrecMin	37.5	0.4	0.6	1.0	1.2	1.6	1.7	1.5	0.6	0.2	0.2	0.8	0.2
	PrecStd	6.4	1.4	1.6	2.9	2.2	2.2	1.2	1.7	1.4	2.7	2.7	2.5	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.4	72.7	78.4	83.0	87.9	94.6	98.9	99.4	92.1	84.8	76.4	66.3
		MCWB	57.5	59.6	61.0	64.9	70.1	73.5	75.1	74.9	72.3	69.5	62.9	58.6
	2%	DB	62.8	66.1	73.9	79.3	84.1	90.5	94.3	95.1	89.5	81.0	71.2	62.0
		MCWB	55.8	55.6	60.0	63.2	69.7	74.1	75.2	74.5	72.6	67.5	59.1	56.4
	5%	DB	57.2	61.2	69.6	75.9	81.6	88.0	91.2	91.5	86.2	77.0	66.7	57.0
		MCWB	50.1	53.3	57.1	61.5	68.7	73.2	75.3	74.2	71.4	64.6	57.5	50.9
	10%	DB	51.9	56.4	64.9	72.6	79.1	85.4	89.2	88.6	82.4	73.0	63.3	52.9
		MCWB	45.1	49.0	54.8	60.3	67.6	72.5	75.1	73.8	69.0	62.8	55.8	47.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.2	62.2	65.0	68.5	74.3	78.1	79.6	79.0	75.7	72.8	66.0	61.9
		MCDB	64.7	70.1	74.7	78.3	83.1	88.9	92.0	90.6	86.4	81.4	72.4	64.7
	2%	WB	57.8	59.0	62.2	66.6	72.4	76.4	78.3	77.5	74.2	70.4	62.7	57.3
		MCDB	61.1	64.0	70.6	75.0	80.7	86.5	89.7	89.2	84.3	77.5	67.2	60.8
	5%	WB	51.4	55.0	59.9	64.5	70.8	75.0	77.3	76.3	73.0	67.9	59.9	52.7
		MCDB	55.9	59.6	66.5	71.8	78.7	84.6	88.1	87.4	82.3	73.5	65.2	55.5
	10%	WB	45.5	49.5	57.0	62.5	69.1	73.8	76.2	75.0	71.7	65.1	56.2	48.3
		MCDB	51.1	56.1	64.0	69.8	76.3	82.6	86.0	85.2	80.0	71.2	62.1	51.9
Mean Daily Temperature Range	5% DB	MDBR	18.9	19.4	21.0	21.4	19.7	19.4	19.5	20.6	21.9	21.4	20.2	17.7
		MCDBR	25.1	25.8	25.7	24.7	22.0	21.9	22.0	24.4	24.0	24.0	24.7	22.3
		MCWBR	17.7	17.4	14.5	12.9	10.8	9.0	7.9	8.3	9.7	12.4	15.4	16.3
	5% WB	MCDBR	21.7	22.7	21.4	20.3	18.5	18.7	19.5	20.5	19.8	19.8	20.4	19.1
		MCWBR	18.4	17.6	14.3	12.5	10.1	9.2	8.3	8.4	8.7	11.7	15.7	16.8
Clear-Sky Solar Irradiance	taub	0.301	0.305	0.337	0.377	0.409	0.423	0.454	0.449	0.385	0.343	0.313	0.296	
	taud	2.487	2.469	2.426	2.317	2.294	2.270	2.230	2.223	2.385	2.487	2.519	2.513	
	Ebn at Noon	284	297	295	287	278	273	263	262	275	279	277	278	
	Edh at Noon	26	30	34	40	42	43	44	43	35	29	25	23	
All-Sky Solar Radiation	RadAvg	749	939	1271	1623	1798	2057	1984	1793	1538	1146	810	633	
	RadStd	70	113	96	104	159	141	116	106	108	131	87	76	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+241
Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page